CE213 Artificial Intelligence Assignment 1

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This assignment is worth 20% of the marks for the module
SUBMISSION DUE ON FASER IN WEEK 24

THE PROBLEM AND TASKS:

For this assignment you are asked to modify and extend a provided Python program using minimax search for playing the Tic-Tac-Toe game on a 3x3 grid to a program for playing the Tic-Tac-Toe game on a 5x5 grid. The winning criterion for playing Tic-Tac-Toe on a 5x5 grid is '5 in a line' (row, column, or diagonal). Tic-Tac-Toe is also called Noughts and Crosses. For details about the game, visit: https://en.wikipedia.org/wiki/Tic-tac-toe.

The provided code is the one for Lab Exercise 2 (Python version), which is available at https://moodle.essex.ac.uk/course/view.php?id=3651&section=17. Before doing this assignment, you are supposed to have run and understood the provided code, which implements minimax search that always returns values from endgame states. Your modified code needs to implement minimax search that can return values of game states at certain depth rather than endgame states by using a heuristic function for evaluating non-endgame states, because returning values from endgame states on a 5x5 grid would be unrealistic - too slow for the minimax search to return values for making a move.

Specifically, your main tasks are as follows:
1) To extend the Board class for playing Tic-Tac-Toe on a 5x5 grid, including its relevant methods;
2) To add a method for calculating the heuristic value of any given game state (board), which could be added in the Board class or the AI_Player class;
3) To modify the minimax method for returning values of non-endgame states rather than endgame states if the depth is too large (larger than a preset maximum depth, say, 6).

It is optional to use alpha-beta pruning in this assignment in the sense that it will not affect the marks for correctness, but you may get a few extra marks from this for quality of your program in terms of its speed of making good moves.

You need to comment the modified and extended code, demonstrating your understanding of the extended code. The general guidelines for code commenting are as follows: There are two types of comments: documentation comments and implementation comments. Documentation comments describe the semantics of a class or method. Good documentation comments should allow someone to use the class and its methods without having to read any source code. In contrast, implementation comments are used to clarify how a particular piece of code operates. You write implementation comments when you feel they are necessary (e.g., for this assignment, it is necessary to comment in detail on the calculation of heuristic value of a game state (board) and important steps in the minimax search method).

SUBMISSION:

The assignment should be submitted through the online coursework submission system (FASER).

You should submit a single zip file with your registration number as the file name, containing all the source code files that comprise your modified and extended program for playing Tic-Tac-Toe on a 5x5 grid, with sufficient comments inside the code.

ASSESSMENT CRITERIA:

Criteria

	Mark	Comments
Correctness of the program (50%)	/50	
Quality of the program (quality of the AI player's moves, efficiency/speed, etc.) (30%)	/30	
Clarity of the program (its structure, etc.) and code commenting (20%)	/20	
Total	/100	

N.B. For each criterion, the following three parts of your work will equally make contributions to the marks you can get:

1) Extending the Board class for playing Tic-Tac-Toe on a 5x5 grid, including its relevant methods
2) Adding a method for calculating the heuristic value of any given game state (board), which could be added in the Board class or the AI_Player class
3) Modifying the minimax method for returning values of non-endgame states rather than endgame states if the depth is too large (larger than a preset maximum depth).

LATE SUBMISSION AND PLAGIARISM

Please refer to the Undergraduate Students' Handbook for details of the School policy regarding late submission and University regulations regarding academic offences:
https://www1.essex.ac.uk/students/exams-and-coursework/academic-offences.aspx

This is an individual assignment. Please acknowledge clearly by commenting your program if any part of your program involves any contribution or help from your classmates or from Internet resources. Otherwise, use of other people's work in your assignment without appropriate acknowledgement and reference will be reported as an SAO case.

As this assignment is about modifying and extending the provided code. You are allowed to use any parts of the code provided for Lab Exercise 2 at Your Own Time, without the need of acknowledgement.

If any problems arise in the assignment, you can discuss them with me via e-mail (**v.k.singh@essex.ac.uk**) or during the CE213 Lab sessions.